

Public Economics

Lecture 9 · Cost-Benefit Analysis

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Today's Questions

- Once we know a public good is underprovided, how does a government decide whether a *specific* project is worth building?
- How do we put a dollar value on things that don't have a market price — saved time, a cleaner river, even a human life?
- How far into the future should we count costs and benefits, and at what rate should we discount them?

Based on

Saez, E., *Econ 131* lecture notes (UC Berkeley); Gruber, J. *Public Finance and Public Policy*, Ch. 8.

8.1 Measuring the Costs of Public Projects

8.2 Measuring the Benefits of Public Projects

8.3 Putting It All Together

What Is Cost-Benefit Analysis?

Cost-benefit analysis

The comparison of the costs and benefits of a public project, used to decide whether it should be undertaken.

Sounds like simple accounting — add up the costs, add up the benefits, compare. In practice it is genuinely hard, because:

- Many relevant resources are **not** priced in a competitive market.
- Some benefits (a life saved, an hour of time, a species preserved) have **no market price at all**.
- Costs and benefits often arrive at very different **points in time**.

8.1 Measuring the Costs of Public Projects

Cash-Flow Accounting and Opportunity Cost

Cash-flow accounting

Calculates costs by simply adding up what government pays for inputs, and benefits by adding up the income or revenue a project generates.

Opportunity cost

The social marginal cost of any resource is the value of that resource in its **next best use**.

The general rule: economic costs are only those costs that come from **diverting** a resource away from its next best use — not simply what the government happens to pay for it.

Measuring Social Cost in Practice

Perfectly competitive markets: Social cost = price. True for most labor and material inputs.

Imperfectly competitive markets:

- **Monopoly** (e.g., asphalt supplied by a monopolist): price = marginal cost + monopoly markup > marginal cost. On efficiency grounds, **social cost** = **marginal cost** — the markup is just a transfer from taxpayers to the monopolist, which matters for distribution but not efficiency.
- **Labor market with unemployment:** suppose a \$10 minimum wage creates involuntary unemployment, and the unemployed would work for \$6 on average. A government project hires them at \$10/hour — but the workers gain a \$4/hour surplus. **Social cost** = $\$10 - \$4 = \$6$, not \$10.

Measuring Future Costs: Present Discounted Value

Present discounted value (PDV)

A dollar next year is worth $1/(1+r)$ of a dollar today, since that dollar could otherwise earn interest r .

A maintenance cost of F paid forever has PDV:

$$\frac{F}{1+r} + \frac{F}{(1+r)^2} + \frac{F}{(1+r)^3} + \cdots = \frac{F}{r}$$

Government typically discounts at its own borrowing rate r (e.g., 6% nominal, 3% real). This is reasonable for **short-run** projects — but becomes problematic over the long run: at $r = 3\%$, \$1 in 100 years is worth only \$0.052 today. Long-run costs like climate change get discounted almost to nothing.

Long-Run Social Discounting: Two Justifications

Social discount rate

The appropriate rate r for computing the PDV of a *social* investment.

Two distinct reasons to discount a future dollar relative to a dollar today:

1. **Absolute (“pure time”) discounting:** people simply prefer \$1 now to \$1 later. On ethical grounds, it's not obvious we should discount *future generations'* welfare this way — except for genuine existential risk (e.g., a meteorite ending civilization).
2. **Economic growth:** future generations will likely be richer, so an extra dollar is worth less to them at the margin — this alone justifies some discounting even with zero pure time preference.

In an ideal world, both effects are already embedded in the market interest rate r , so using today's r would be enough.

Weitzman (1998): Discount the Distant Future at Its Lowest Rate

The problem: we don't actually know how growth — and hence r — will evolve over the next century. If the economy collapses (e.g., from unchecked climate change), future generations will be **poor**, and we should *not* discount their welfare heavily.

A simple illustration

50% chance of zero growth ($r = 0\%$): \$1 in 100 years = \$1 today.

50% chance of normal growth ($r = 3\%$): \$1 in 100 years = \$0.052 today.

Expected value today: $0.5(1) + 0.5(0.052) = \$0.552$.

Solving $(1 + \bar{r})^{-100} = 0.552$ gives an implied discount rate $\bar{r} \approx 0.6\%$ — far below either scenario's own rate.

Takeaway

When there's a real chance growth stops, the distant future should be discounted at a **low** rate.

8.2 Measuring the Benefits of Public Projects

Valuing Time: Using Wages

If individuals freely optimize their labor supply, they're indifferent between working one more hour and enjoying one more hour of leisure at the margin.

Hourly wage = value of one hour of leisure

So a project that saves people time (e.g., faster commutes) can be valued using wages — whether people use the saved time to work more or relax more. **Problems in**

practice:

1. Individuals often can't freely trade off hours and leisure — many jobs come with fixed-hours constraints.
2. An hour *stuck in traffic* is worse than an hour of ordinary leisure lost — so the true value of reducing congestion exceeds the wage-based estimate.

Valuing Time: Contingent Valuation

Contingent valuation

Directly asking individuals to value an option they are not currently choosing, or cannot choose at all.

Often the **only** feasible method when there is no market price whatsoever:

- The value of saving an endangered species.
- The value of keeping the Arctic pristine.

Popular among environmental advocates as a way to argue that a cause is “worth” a large number — which is exactly where the method’s problems begin.

The Problems with Contingent Valuation

Diamond & Hausman show that the structure of a contingent-valuation survey can swing the answer wildly:

- **No real cost to respondents** — easy to exaggerate value when you don't actually have to pay.
- **Isolation effects:** asking about one cause alone vs. many causes together changes the stated value.
- **Order effects:** which question comes first matters.
- **Embedding effect:** people state a similar willingness to pay for saving *one* lake as for saving *three* lakes.

The right comparison

A rational allocation requires looking at *all* competing causes together — weighing one cause in isolation is not a sound basis for public policy. Government, which must set a single budget across all of them, is best placed to make that trade-off.

Valuing Time: Revealed Preference

Revealed preference (hedonic approach)

Let people's actual choices reveal their valuation.

- **1970s US gas price controls** created queues at most stations — but small independent stations, exempt from the controls, could charge more and had shorter queues. Comparing the price gap to the queue-length gap: saving \$10 by queuing one hour $\Rightarrow$ one hour is worth about \$10.
- **FasTrak toll lanes** (San Francisco Bay Area): what drivers pay to skip congestion reveals the value of time *and* of avoiding the unpleasantness of queuing itself.

These estimates capture the value of time for the *marginal* person — the one just indifferent between paying and waiting — not necessarily the average person.

Valuing human life is the single hardest, most ethically fraught issue in cost-benefit analysis.

- Yet virtually *every* government expenditure changes the odds of saving a life somewhere — safer roads, health care spending, and so on.
- So some value must be placed on a **statistical** life — a small change in the probability of death across a large population.

A crucial distinction

We can meaningfully price a **statistical** life (fewer expected accidents across a population) even though pricing a specific, identified, **real** life is neither possible nor something anyone would attempt.

Valuing Saved Lives: Revealed Preference

As with time, economists' preferred method is revealed preference: how much are people actually willing to pay to reduce their own odds of dying?

Compensating differentials

Extra (or reduced) wages paid to compensate workers for a job's negative (or positive) amenities — including mortality risk.

Example: a \$10,000 enlistment bonus was needed to recruit soldiers during the Afghanistan/Iraq wars, which carried an extra 1/1,000 chance of dying relative to peacetime.

$$\text{Value of a statistical life} = \frac{\$10,000}{1/1,000} = \$10 \text{ million}$$

Limitations: requires rational behavior, and again values the *marginal* person's life, not the average.

Source: US studies put the revealed value of a statistical life at about \$9.3 million on average today.

Application: Speed Limits and the Value of a Statistical Life

Ashenfelter & Greenstone (2004), *JPE*: in 1987 the US federal government allowed states to raise rural highway speed limits from 55 to 65 mph — 21 states did so.

- The higher limit raised speeds by $\approx 3.5\%$ and raised fatality rates by $\approx 35\%$.
- That trade-off implies about **125,000 hours** of travel time saved per additional life lost.
- Valuing time saved at the average hourly wage: adopting states were effectively willing to accept a fatality risk in exchange for saving **\$1.54 million** (1997 dollars) per statistical life.

Interpretation

These states were revealing a value of a statistical life of **at most** \$1.54 million — well below the \$9–10 million found in other contexts, a useful caution about how much these estimates vary by setting.

8.3 Putting It All Together

Common Counting Mistakes

- **Counting secondary benefits:** a new highway boosts commerce nearby — but often just *relocates* activity from somewhere else rather than creating it.
- **Counting labor as a benefit:** jobs “created” by a project are a **cost**, not a benefit — those workers are not producing something else while employed on this project.
- **Double-counting benefits:** a project that reduces commuting time *and* raises nearby house values is often counting the **same** benefit twice, since the house-price gain largely capitalizes the time savings.

Distributional Concerns and Uncertainty

Distributional concerns

The costs and benefits of a public project rarely fall on the same people. A project can pass a cost-benefit test in aggregate while still making some groups clearly worse off.

Uncertainty

Both the costs and the benefits of public projects are often highly uncertain — construction costs run over budget, and benefits (time saved, lives saved, growth effects) depend on assumptions that may not hold.

A cost-benefit number is a starting point for a policy debate, not a verdict that ends it.

Ashenfelter, O. & Greenstone, M. (2004), *AER* 94(2).

Diamond, P. & Hausman, J. (1994), *JEP* 8(4).
Gruber, J., *Public Finance and Public Policy*, Ch. 8.

Saez, E., *Econ 131* lecture notes, UC Berkeley.
Weitzman, M. (1998), *JEEM* 36(3).

Thank you!

Questions and discussion